Trevor Werbin

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Education

University of Waterloo - B.A.Sc Mechanical Engineering, Mechatronics Option **Sep 2025 - Apr 2030**

- Electromechanical Devices and Power Processing, Properties of Materials, Mechanics of Deformable Solids, Adv. Calculus
- Member of the Waterloo Space Rover Design Team (University of Waterloo Robotics Team)

Work Experience

Ciena Corporation **Ottawa, Ontario**

Mechanical Engineering Intern (R&D) **May - Aug 2026**

- Created a press-fit alignment jig using PTC Creo to cut PCB assembly placement errors by 15%
- Designed & executed a 17 configuration DOE to characterize product thermal performance for a large customer
- Increased production 3D printer uptime by 20% and delivered multiple >24hr prints on two $200k 3D printers
- Implemented a filament inventory tracking system that cut print costs by $100/mo and reduced filament waste by 5%
- Reduced 3D printer setup and changeover times by 50%, increasing production volume by 10%, from designing modular, custom-fitted tool organizers for Hylo AON3D printer using PTC Creo and press-fit techniques
- Saved 3hrs/week of engineering labor by personally fabricating and assembling prototype products for internal testing using machine tools and 3D printing
- Supported accurate manufacturing of 25 parts by creating and releasing toleranced and DFM reviewed drawings (GD&T) for production via PLM
- Improved product build by designing, updating and replacing parts in 50+ assemblies with PTC Creo and Windchill

Ontario Public Service, Ministry of Health, Ministry of Children **Toronto, Ontario**

Administrative Coordinator, Director's Office (part-time) **Jul 2024 - Aug 2025**

- Defined 7 process success KPIs & executed risk assessments for government project approval system
- Established knowledge base of project and human resource information using SharePoint

Projects (trevorwerbin.github.io/mechanical-portfolio/)

Autonomous Cookie Icing Machine

SolidWorks, C++, Fabrication, DFM, Circuits, Machining Safety, AutoCAD

- Designed an autonomous cookie icing machine in 3 months using laser cutting, 3D printing and Vex IQ components
- Wired and integrated 4 motors and 5 sensors, applying the engineering design process to develop and validate the multi-actuator electromechanical system
- Integrated user input front-end with motion control backend to convert custom designs into machine toolpaths

Mechanical Thumb Toy (ME100 Final Project)

SolidWorks, 3D Printing, Iterative Design, Functional Testing

- Designed, printed and tested a wearable actuating thumb meeting a 30N grip force spec with <1 s actuation
- Iterated 3 designs through functional testing to achieve design specs

Solar Car Canopy Design (UW Midnight Sun Team)

SolidWorks, Surface Modelling, Lofting, Split Lines

- Reduced overall wind resistance by designing an aerodynamic canopy for the solar car

DIY Dynamometer (ME100 Project)

Excel, Data Analysis, Technical Report Writing

- Quantified a 45% deviation from manufacturer torque curves using a custom built rope-brake dynamometer
- Created torque-speed and elastic force-displacement graphs in Excel to compare experiment vs. manufacturer data

Technical & Business Skills

Tools: Parametric CAD (PTC Creo, Solidworks), AutoCAD, Agile PLM, Machining, Additive Manufacturing (FDM, SLA)
Programming: MATLAB, C#, Python, C++
Business Skills: Project Management, Excel, Google Sheets, Data Analysis, MS SharePoint, Presentations & Reports